Assessment Menu: Teacher Guide

The Inequalities Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
SOLVE	Solve inequalities	A1, A2, A3, E1, E2
COMPOUND	Compound inequalities	A5, E3, E4, S2
REASON	Graph and reason	A4, E6, S3
APPLY	Write and use inequalities in context	A6, E5, S1, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 SOLVE	Solve: $x + 8 > 3$	**Answer:** $x > -5$ Subtract 8 from both sides: $x > 3 - 8$ $x > -5$	3 pts: Subtracts 8 from both sides 2 pts: Correct answer $x > -5$ with the correct sign
A2 SOLVE	Solve: $-3x \le 12$	**Answer:** $x \ge -4$ Divide both sides by -3 and **flip** the sign: $x \ge \frac{12}{-3}$ $x \ge -4$	2 pts: Divides both sides by -3 1 pts: Flips the inequality sign 2 pts: Correct answer $x \ge -4$
A3 SOLVE	Solve: $2x - 5 < 9$	**Answer:** $x < 7$ Add 5: $2x < 14$ Divide by 2: $x < 7$	2 pts: Adds 5 to both sides 1 pts: Divides by 2 2 pts: Correct answer $x < 7$
A4 REASON	Graph the solution of $x \ge -2$ on the number line.	**Answer:** Closed circle at -2 with shading and an arrow to the right. The "≥" sign includes -2, so use a **closed** circle at -2. Shade to the right, toward the larger numbers, and end with an arrow.	2 pts: Closed circle at -2 2 pts: Shades to the right 1 pts: Arrow showing the solution continues
A5 COMPOUND	Solve: $-3 < x - 2 \le 4$	**Answer:** $-1 < x \le 6$ Add 2 to all three parts: $-3 + 2 < x \le 4 + 2$ $-1 < x \le 6$	2 pts: Adds 2 to all three parts 3 pts: Correct answer $-1 < x \le 6$ (both symbols correct)
A6 APPLY	Movie tickets cost $9 each. Write an inequality for the number of tickets *t* Sam can buy with $50, solve it, and say what the answer means.	**Answer:** $9t \le 50$, $t \le \frac{50}{9}$ (about 5.6): Sam can buy at most 5 tickets. $9t \le 50$ $t \le \frac{50}{9} \approx 5.6$ Tickets come in whole numbers, so Sam can buy **at most 5** tickets.	2 pts: Inequality $9t \le 50$ (≤, not =) 2 pts: Solves correctly: $t \le \frac{50}{9}$ or about 5.6 1 pts: Interprets: at most 5 whole tickets

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 SOLVE	Solve: $5 - 2(x + 1) \geq 3x + 8$	**Answer:** $x \leq -1$ Distribute: $5 - 2x - 2 \geq 3x + 8$ Combine: $3 - 2x \geq 3x + 8$ Add $2x$ and subtract 8: $-5 \geq 5x$ Divide by 5: $-1 \geq x$, so $x \leq -1$	2 pts: Distributes the -2 correctly 2 pts: Combines like terms 3 pts: Gets the variable alone with correct steps 3 pts: Correct answer $x \leq -1$ with the sign pointing the right way
E2 SOLVE	Solve: $\frac{x - 4}{-3} < 2$	**Answer:** $x > -2$ Multiply both sides by -3 and **flip** the sign: $x - 4 > -6$ Add 4: $x > -2$	3 pts: Multiplies both sides by -3 2 pts: Flips the sign 2 pts: Adds 4 correctly 3 pts: Correct answer $x > -2$
E3 COMPOUND	Solve and graph: $-5 \leq 3x + 4 < 19$	**Answer:** $-3 \leq x < 5$ Subtract 4 from all parts: $-9 \leq 3x < 15$ Divide all parts by 3: $-3 \leq x < 5$ Graph: closed circle at -3, open circle at 5, shade between.	2 pts: Subtracts 4 from all three parts 2 pts: Divides all three parts by 3 2 pts: Correct answer $-3 \leq x < 5$ 4 pts: Graph: closed circle at -3, open circle at 5, shading between (1 pt each, plus 1 for correct endpoints)
E4 COMPOUND	Solve and graph: $2x - 1 \leq -7$ **or** $4x + 3 > 15$	**Answer:** $x \leq -3$ or $x > 3$ First: $2x \leq -6$, so $x \leq -3$ Second: $4x > 12$, so $x > 3$ Graph: closed circle at -3 shaded left, open circle at 3 shaded right.	3 pts: Solves the first inequality correctly 3 pts: Solves the second inequality correctly 1 pts: Uses “or” correctly in the answer 3 pts: Graph: closed circle at -3 shaded left, open circle at 3 shaded right
E5 APPLY	A gym charges a $25 joining fee plus $12 per month. Maya can spend at most $145. Write and solve an inequality for the number of months *m* she can belong, and state the answer in words.	**Answer:** $25 + 12m \leq 145$, so $m \leq 10$: Maya can belong for at most 10 months. $25 + 12m \leq 145$ Subtract 25: $12m \leq 120$ Divide by 12: $m \leq 10$ Maya can belong for **at most 10 months**.	3 pts: Correct inequality $25 + 12m \leq 145$ 3 pts: Solves correctly (subtract 25, divide by 12) 2 pts: Correct answer $m \leq 10$ 2 pts: States the meaning in words
E6 REASON	Ravi solved $-2x + 6 > 14$ like this: $-2x > 8$ $x > -4$ Find his mistake, explain it, and give the correct solution.	**Answer:** He did not flip the sign when dividing by -2. Correct solution: $x < -4$. His first step ($-2x > 8$) is correct. Dividing both sides by a **negative** number reverses the inequality: $x < -4$. Check: $x = -5$ gives $-2(-5) + 6 = 16 > 14$ ✓, but $x = 0$ (which is > -4) gives 6, not > 14.	4 pts: Identifies the mistake: the sign must flip when dividing by a negative 3 pts: Explains why in a complete sentence 3 pts: Correct solution $x < -4$

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 APPLY	A taxi charges $4 plus $3 per mile. Jordan wants to spend **at least $13 and at most $28**. Write a compound inequality for the number of miles *m*, solve it, and describe the trips Jordan can take.	**Answer:** $13 \le 4 + 3m \le 28$, so $3 \le m \le 8$: trips of 3 to 8 miles. $13 \le 4 + 3m \le 28$ Subtract 4 from all parts: $9 \le 3m \le 24$ Divide all parts by 3: $3 \le m \le 8$ Jordan can take trips of **3 to 8 miles**.	6 pts: Correct compound inequality $13 \le 4 + 3m \le 28$ (3 pts for each correct bound) 6 pts: Solves all three parts correctly 4 pts: Correct answer $3 \le m \le 8$ 4 pts: Describes the trips in words (3 to 8 miles, endpoints included)
S2 COMPOUND	Solve and graph: $-4 < \frac{5 - 3x}{2} \le 7$	**Answer:** $-3 \le x < \frac{13}{3}$ Multiply all parts by 2: $-8 < 5 - 3x \le 14$ Subtract 5 from all parts: $-13 < -3x \le 9$ Divide all parts by -3 and **flip both signs**: $\frac{13}{3} > x \ge -3$ Written smallest to largest: $-3 \le x < \frac{13}{3}$ (about 4.33)	4 pts: Multiplies all three parts by 2 4 pts: Subtracts 5 from all three parts 4 pts: Divides by -3 and flips both signs 4 pts: Correct answer $-3 \le x < 13/3$ written smallest to largest (2 pts for bounds, 2 pts for symbols) 4 pts: Graph: closed circle at -3, open circle at 13/3 (a little past 4), shading between
S3 REASON	The solution of $2x + a > 11$ is $x > 4$. **(a)** Find the value of *a* and show how you know. **(b)** Using that value of *a*, solve $-2x + a > 11$.	**Answer:** (a) $a = 3$ (b) $x < -4$ (a) Solve $2x + a > 11$: $2x > 11 - a$, so $x > \frac{11 - a}{2}$ This must equal 4: $\frac{11 - a}{2} = 4$, so $11 - a = 8$ and **$a = 3$**. (b) $-2x + 3 > 11 \rightarrow -2x > 8 \rightarrow$ divide by -2 and flip: **$x < -4$**.	4 pts: Solves for x in terms of a 4 pts: Sets the boundary equal to 4 2 pts: Correct $a = 3$ 4 pts: (b) Substitutes $a = 3$ and isolates $-2x > 8$ 4 pts: (b) Flips the sign: $x < -4$ 2 pts: Work clearly shown for both parts
S4 APPLY	A greenhouse is 60°F when its heater turns on, and it warms 5°F every hour. Plants grow best when the temperature is **at least 65°F but below 80°F**. Write and solve a compound inequality for the number of hours *h* the heater has been on, then graph the solution.	**Answer:** $65 \le 60 + 5h < 80$, so $1 \le h < 4$. $65 \le 60 + 5h < 80$ Subtract 60 from all parts: $5 \le 5h < 20$ Divide all parts by 5: $1 \le h < 4$ Graph: closed circle at 1, open circle at 4, shade between.	6 pts: Correct compound inequality $65 \le 60 + 5h < 80$ (3 pts for each bound) 4 pts: Subtracts 60 from all parts 4 pts: Divides all parts by 5; correct answer $1 \le h < 4$ 6 pts: Graph: closed circle at 1 (2), open circle at 4 (2), shading between (2)